

1

Outline: Week 02 A: Monday

- I) Background Requirement
- II) Topics
- III) Assessments & Team Project
- III) A Brightspace Tour

Outline: W02A

I) Background Requirement

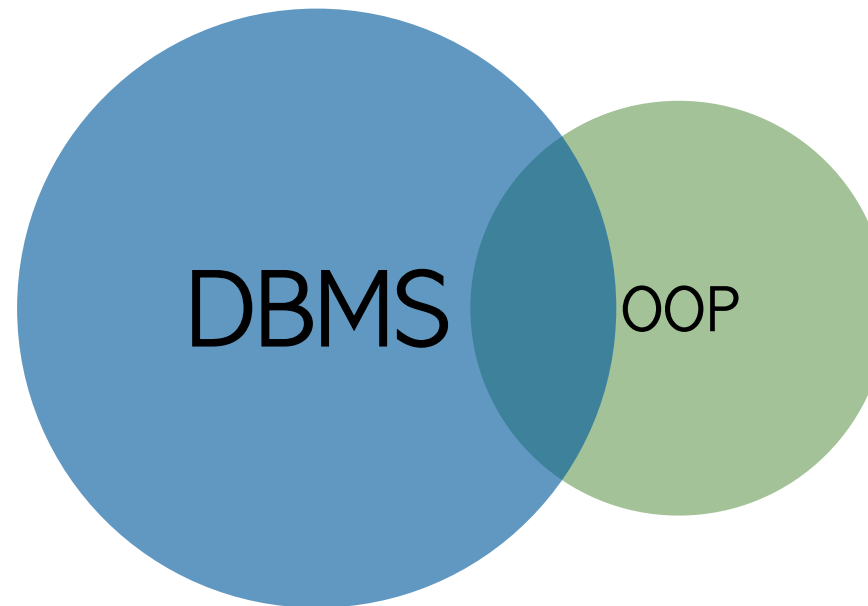
II) Topics

III) Assessments & Team Project

III) A Brightspace Tour

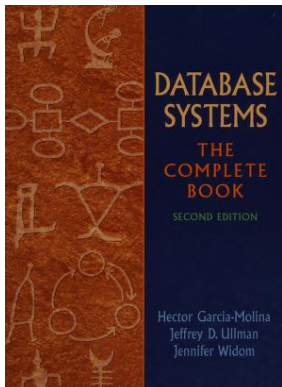
Background

ER Diagram
Relational Algebra
SQL
DBMS e.g., MySQL

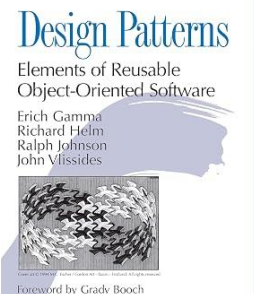


Object-Oriented Programming
Software Layered Architecture

Data Access Layer
Relational to Object Mapping
User Interface



Database Systems: The Complete Book
by Hector Garcia-Molina, Jeffrey Ullman, Jennifer Widom
2008



Design Patterns: Elements of Reusable Object-Oriented Software
by Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides
1994 (Gang of 4 Design Patterns)

Outline: W02A

I) Background Requirement

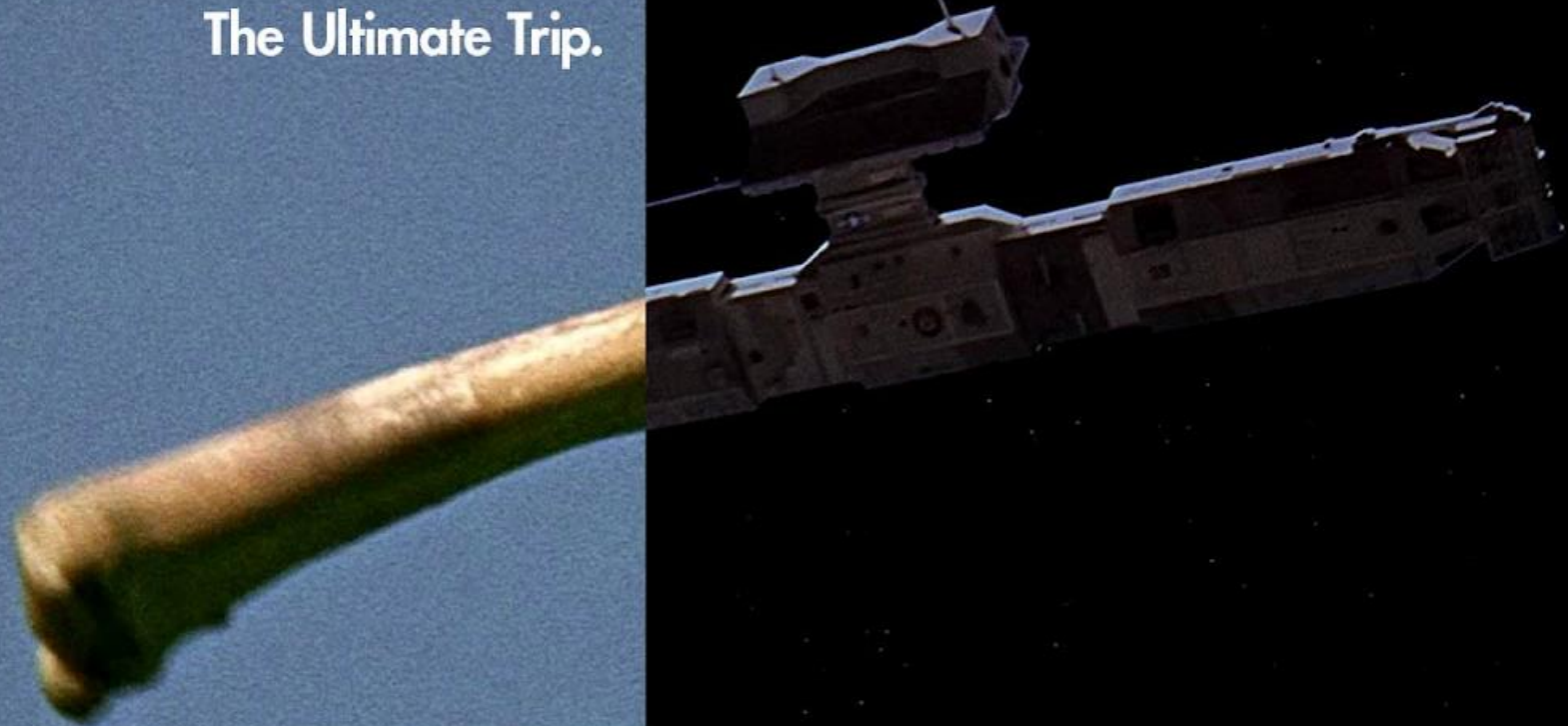
II) Topics

III) Assessments & Team Project

III) A Brightspace Tour

2026: A Data Odyssey

The Ultimate Trip.



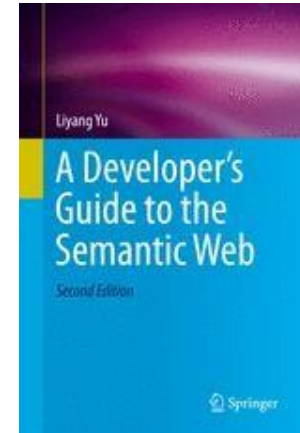
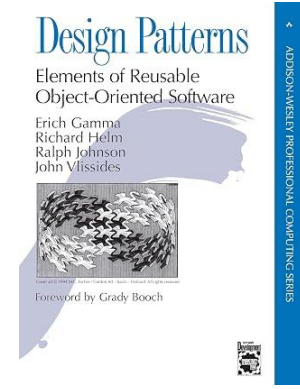
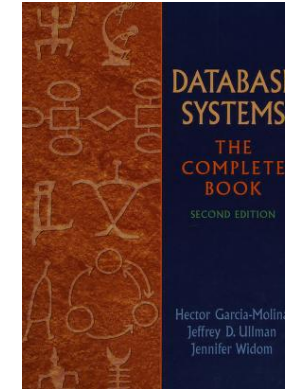
From apelike ancestors to human being: 6 million years
From bones to spaceships

2001: A Space Odyssey (1968), Stanley Kubrick

Topics

- Conceptual Modeling by Entity-Relationship
- Conceptual Modeling by Classes in OOP
- Logical Modeling by Objects in OOP
- Logical Modeling by Relational
- Physical Modeling by SQL
- Logical Modeling by Resource Description Framework (RDF)
- Conceptual Modeling by RDF Schema (RFDS)
- Conceptual Modeling by Web Ontology Language (OWL)
- Logical Modeling by OWL
- Physical Modeling by SPARQL

“sparkle”, a recursive acronym for SPARQL Protocol and RDF Query Language



A Developer's Guide to the Semantic Web,
by Liyang Yu, 2014

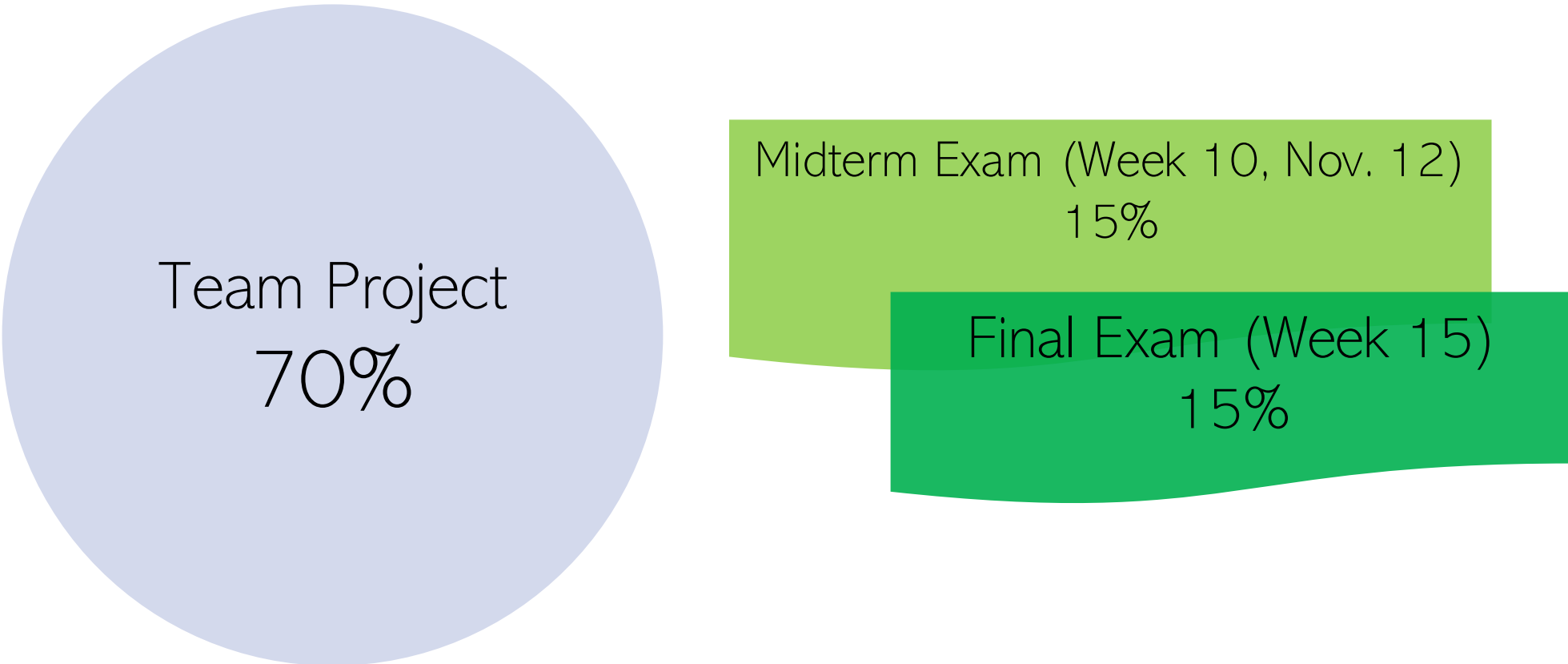
7

Outline: W02A

- I) Background Requirement
- II) Topics
- III) Assessments & Team Project**
- III) A Brightspace Tour

8

Outline: W02A



Team Project
70%

Midterm Exam (Week 10, Nov. 12)
15%

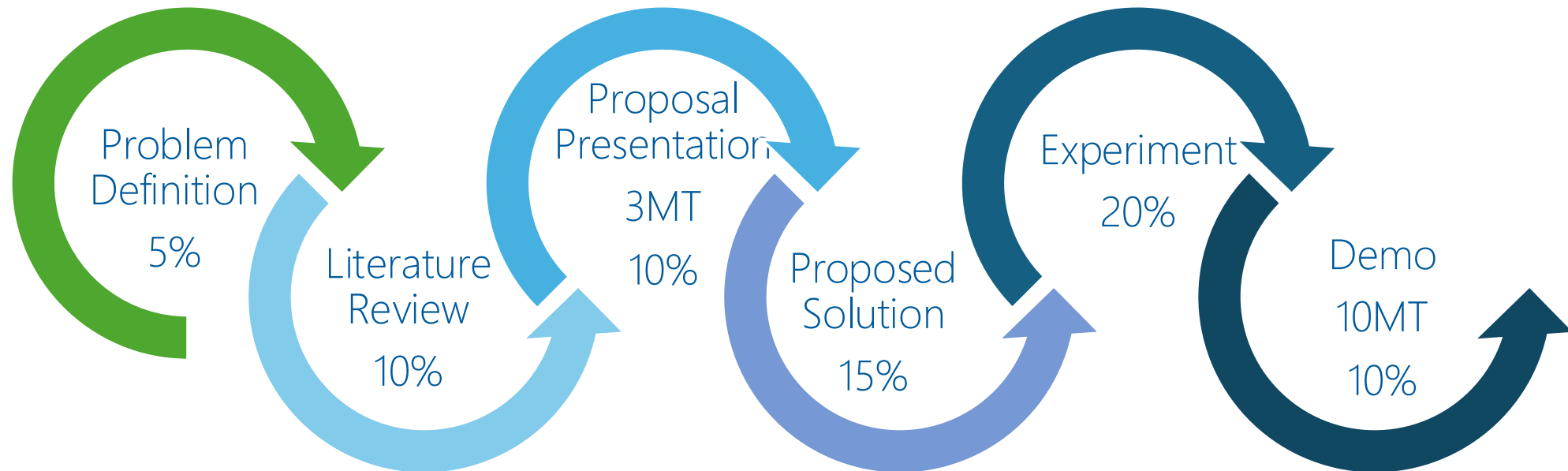
Final Exam (Week 15)
15%

9

Team Project

Min. 2, Max. 3 Members

Course-long Project vs. Final Project

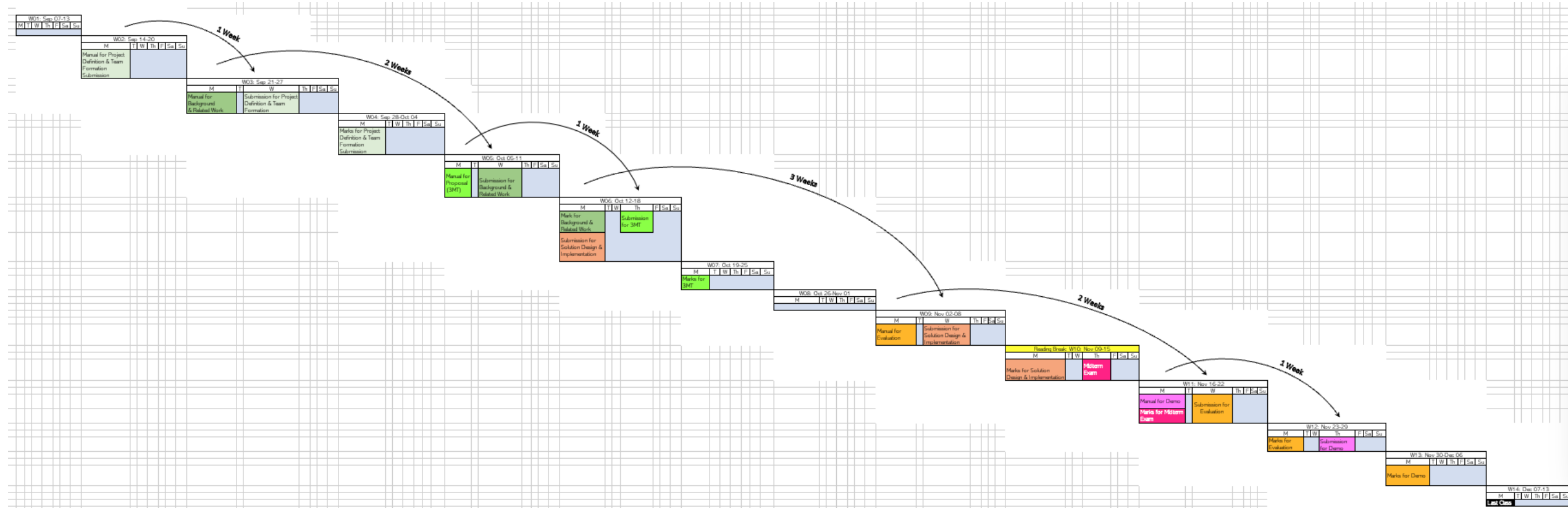


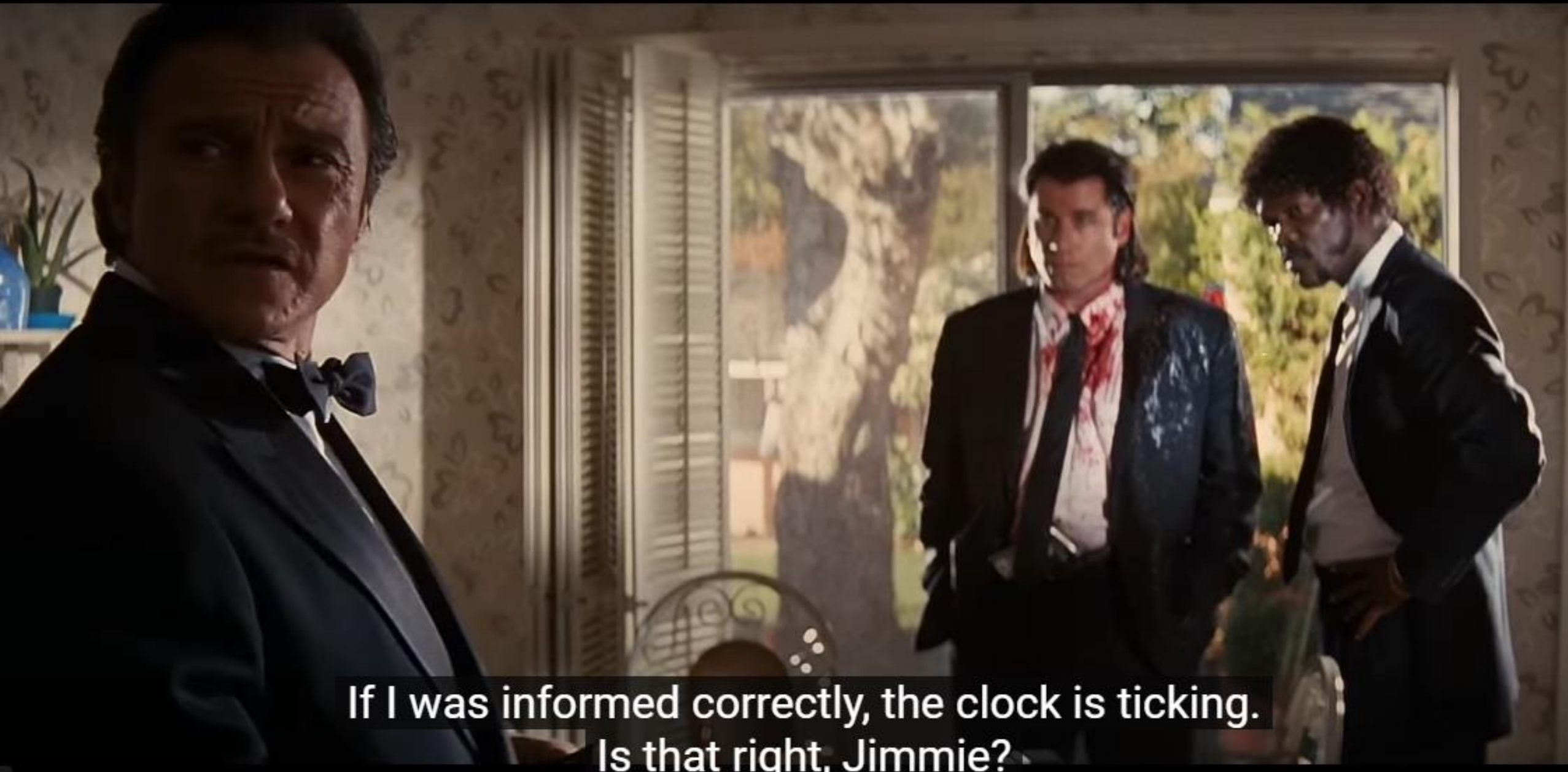
10

Weekly Schedule

MON	TUE	WED	THU	FRI	SAT	SUN	MON
New Submission Manual Prev. Submission Marks Release		Submission Deadline Midnight					
Lecture A: 8:30-9:50 Elliot Building 061: Hossein			Lecture B: 8:30-9:50 Elliot Building 061: Hossein				
Office A: 10:00-11:00 ECS613: Hossein			Office B: 10:00-11:00 ECS613: Hossein				
		Marking					Marking - 12:00 AM
	Monitor Discussion Board						
Link Lecture Video			Link Lecture Video				Release Marks, Stats, and Make Announcer

Timeline





If I was informed correctly, the clock is ticking.
Is that right, Jimmie?

Team Project

Take a Small but Novel Step!

Data Modeling Security and Safety Events

Data Modeling Security and Safety Events in Victoria

Data Modeling Security and Safety Events in University of Victoria

Data Modeling Security and Safety Events in University of Victoria

Team Project

Example: An Ontology for Describing Security Events

https://hosseinfani.github.io/res/papers/2015_SEKE_An_Ontology_for_Describing_Security_Events.pdf

<https://www.slideshare.net/slideshow/an-ontology-for-describing-security-events/238327365>

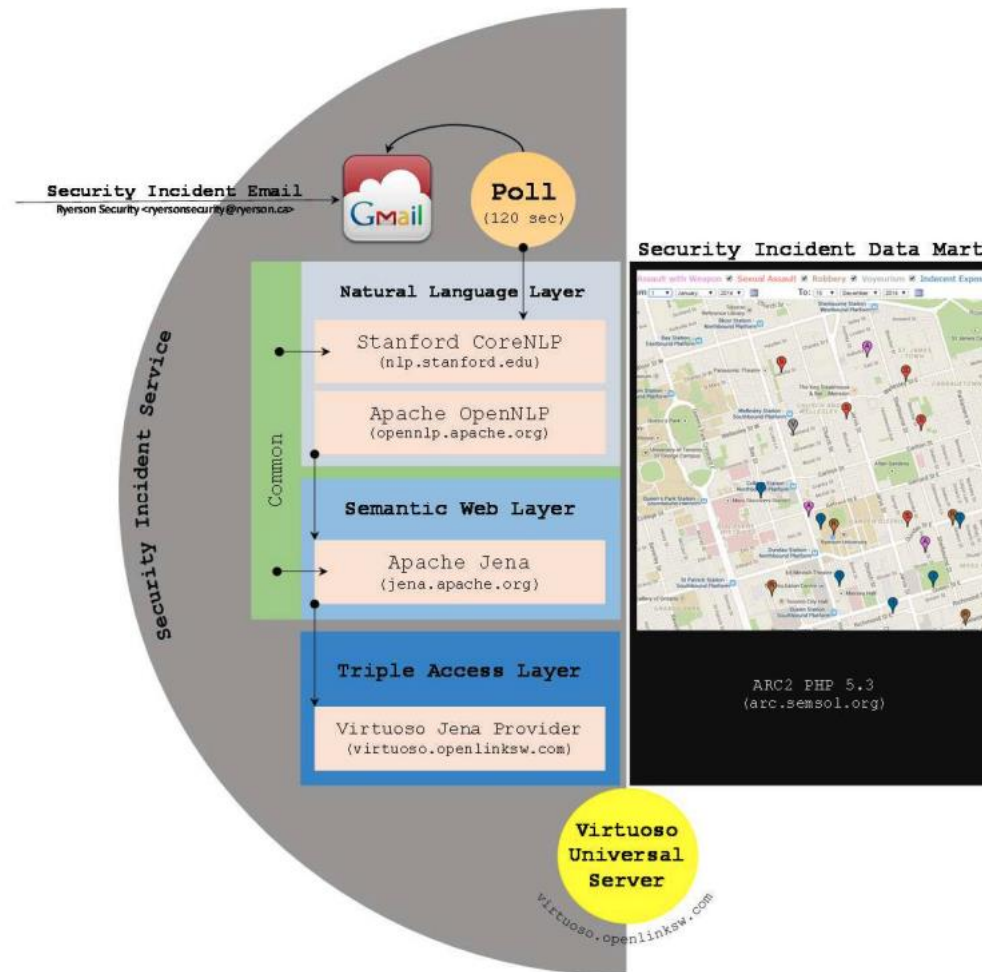


Figure 3. Machine Readable Security Incident Notification software system architecture. The left hemisphere is the software service which polls each 2 minutes a gmail account, seurityincidentsemanticweb@gmail.com, which is subscribed to be notified for any security incident by Ryerson University Integrated Risk Management. After fetching new incidents, the Natural Language Layer parse the text and extract information to an object of type `SecurityIncident`, defined in `Common`, and hand it to next layer. The Semantic Web Layer serialize the populated object to OWL individuals and pass them to Triple Access Layer to store them in the Virtuoso triple store. The right hemisphere is the primitive dashboard at <http://semionet.rnet.ryerson.ca/sio/> which shows the incidents geographical distribution filtered by the date of event.

15 Team Project: GitHub Repo

Branding: Pick a name for your project, design a logo, ...

Structured: Directories, Layers, Output, ...

Contributions: Pull Requests, Commits, ...

`license.txt`

`readme.md`: Explanation, setup, samples, ...

Scripts: For sample run, explaining architecture, ...

Scripts: **Should not be the main codebase!**

Team Project: GitHub Repo

Let's have a look at some good and bad repos

https://github.com/hosseinfani/security_incident_ontology

<https://github.com/fani-lab/OpeNTF>

<https://github.com/fani-lab/LADy>

<https://github.com/fani-lab/Osprey>



Outline: W02A

- I) Background Requirement
- II) Topics
- III) Assessments & Team Project
- III) A Brightspace Tour

18

A Brightspace Tour

 bright.uvic.ca/d2l/home/513675

Fall 2026 CSC 501 A01 A02 X

Hossein Fani
as Learner[Course Home](#) [Content](#) [Classlist](#) [Grades](#) [Class Progress](#) [Course Tools](#) [UVic Resources](#) [Help](#)

CSC501: Algorithm and Data Models

[Outline](#)[Instructor & Assistant](#)[Schedule \(Weekly, Timeline\)](#)[Lectures \(Forum, Slides, Recordings, Hyperlinks, Resources, ...\)](#)[Project \(Forum, Milestones, Manuals, ...\)](#) [Milestone 1: Proposal: Problem Definition & Team Formation](#)[Exams \(Forum, ...\)](#)